AIR BAG SYSTEM [STANDARD DEPLOYMENT CONTROL SYSTEM]

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Outline

- With the air bag system, the SAS control module operates (deploys) the air bags according to the degree of impact during the collision to supplement the occupant protection function of the seat belts.

Warning

- **The air bag system supplements the occupant protection function of the seat belts, however the intended effect of the air bag system cannot be obtained if the seat belts are not fastened correctly. When driving the vehicle, always fasten the seat belts correctly.**

Function

- With the air bag system, the SAS control module calculates the degree of impact based on the collision detection signal from each of the following sensors, and operates (deploys) the required parts.

×: Collision detection
××: Operation (deployment)
—: No detection

Collision area	Sensor			Operation (deployment) part				
	Crash sensor integrated into SAS control module	Crash sensor integrated into crash zone sensor	Crash sensor integrated into side air bag sensor *2	Driver-side air bag module	Passenger-side air bag module	Curtain air bag module (driver-side/ passenger-side) *1	Side air bag module (driver-side/ passenger-side) *2	Front pre-tensioner seat belt (driver-side/ passenger-side)
Collision from front of vehicle or frontal offset	×	—	—	—	—	—	—	—
	—	×	—	—	—	—	—	—
	×	×	—	××	××	××*3	××*3	××
Collision from side of vehicle	×	—	—	—	—	—	—	—
	—	—	×	—	—	—	—	—
	×	—	×	—	—	××*4	××*4	××*4
Collision from rear of vehicle	×	—	—	—	—	—	—	—

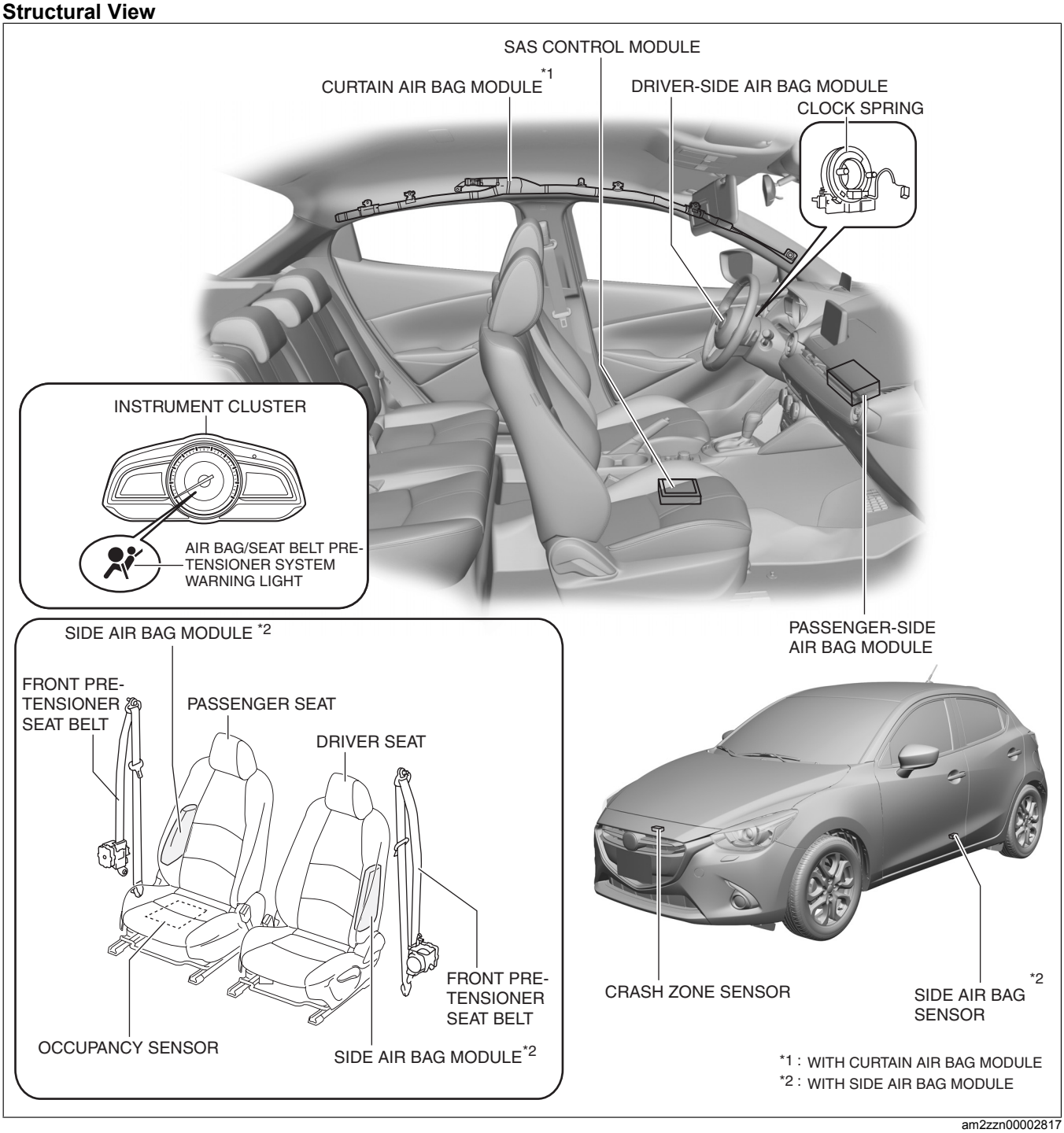
*1 : With curtain air bag module
*2 : With side air bag module
*3 : Operates (deploys) by degree of impact during collision.
*4 : Operates (deploys) on the side the collision is detected.

System conditions display function

- The air bag system detects each part condition and the SAS control module sends the system status signal to the instrument cluster.
- The instrument cluster notifies the user of the system conditions using the following warning light or alarm based on the system condition signal.

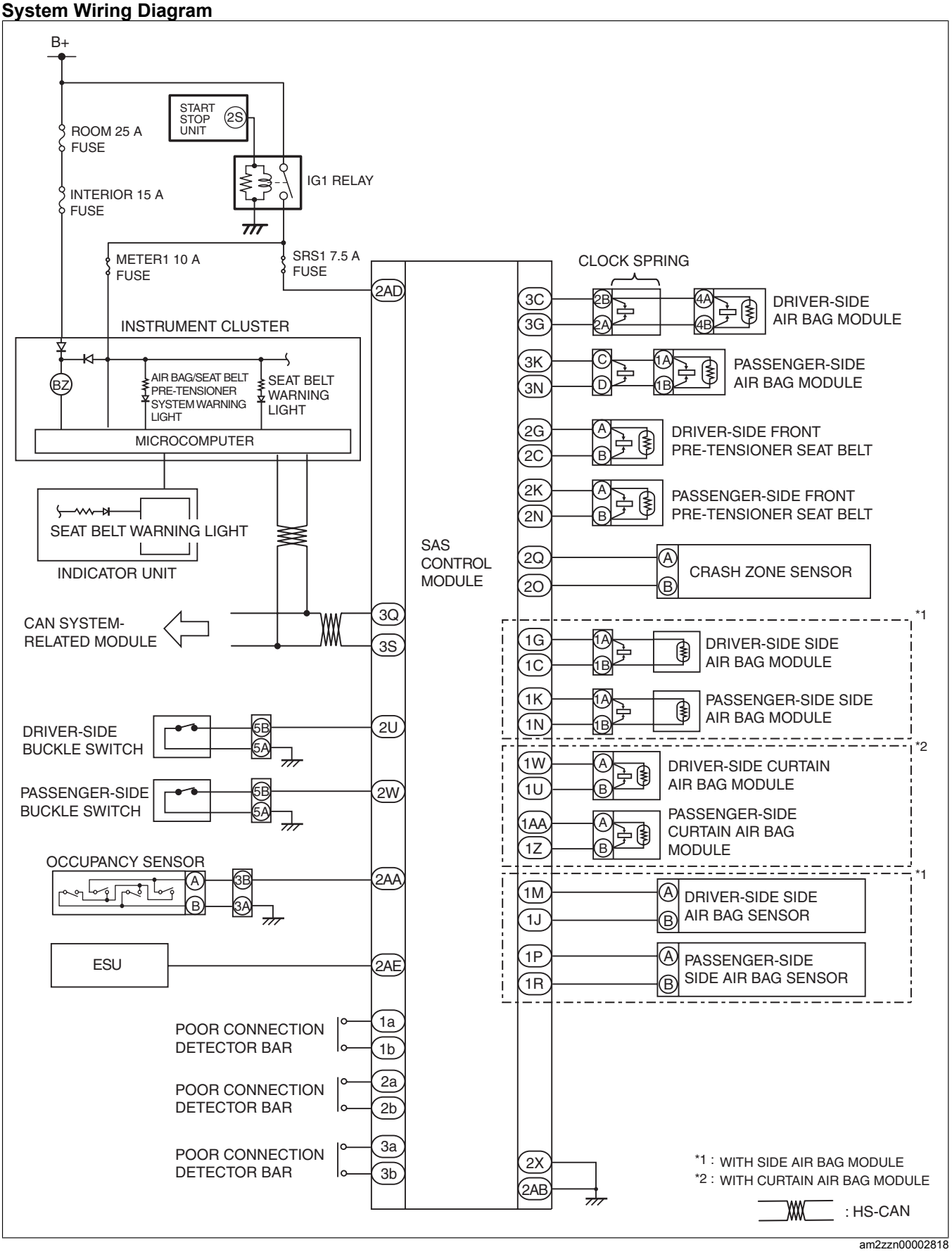
Warning light and alarm	Reference
Air bag/front seat belt pre-tensioner system warning light	(See AIR BAG/SEAT BELT PRE-TENSIONER SYSTEM WARNING LIGHT [STANDARD DEPLOYMENT CONTROL SYSTEM].)
Air bag system warning alarm	(See AIR BAG SYSTEM WARNING ALARM [STANDARD DEPLOYMENT CONTROL SYSTEM].)
Seat belt warning light	(See SEAT BELT WARNING ALARM.) (See INDICATOR UNIT.)

Structural View



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System Wiring Diagram

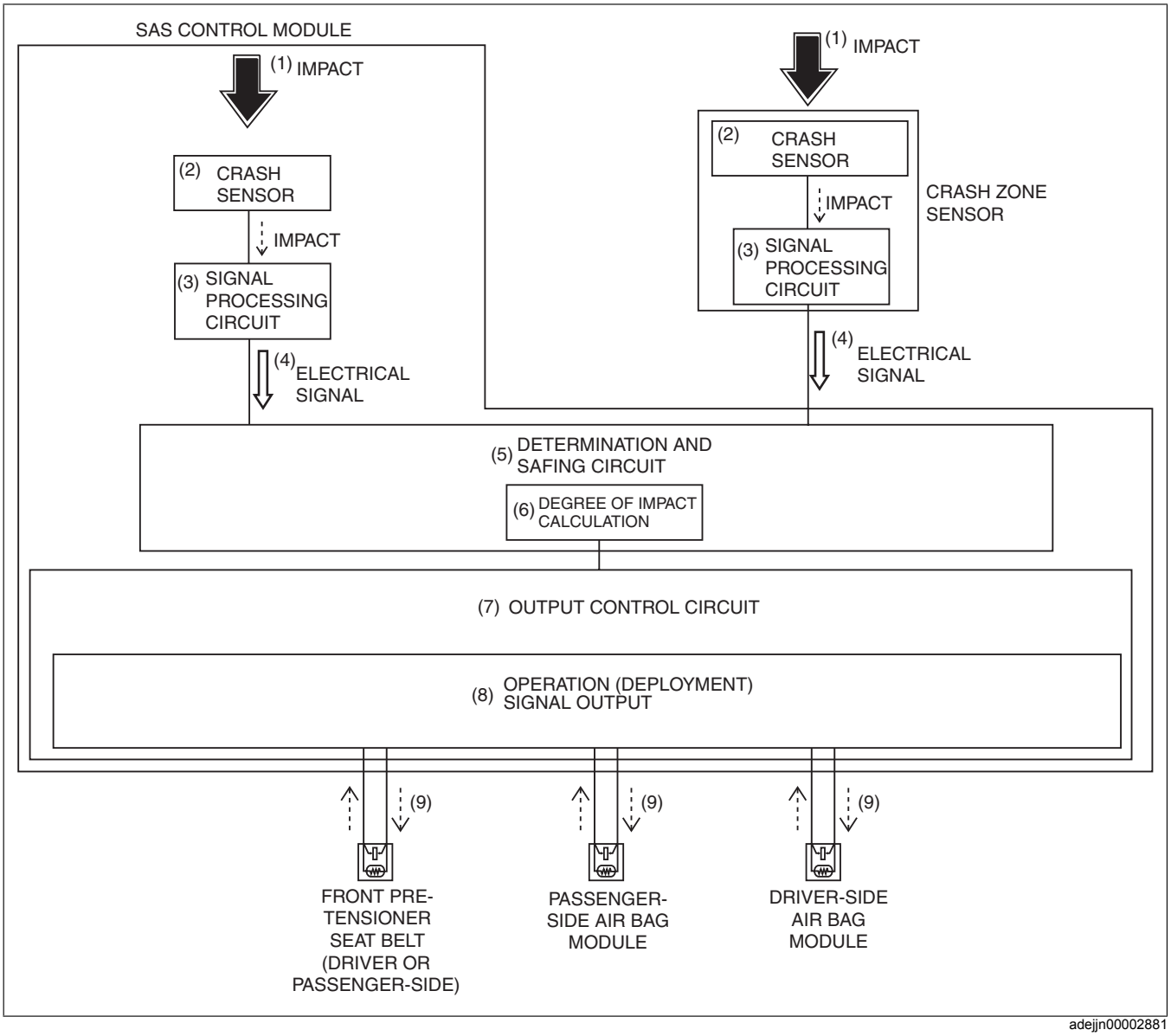


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Operation

Operation during frontal or frontal offset collision

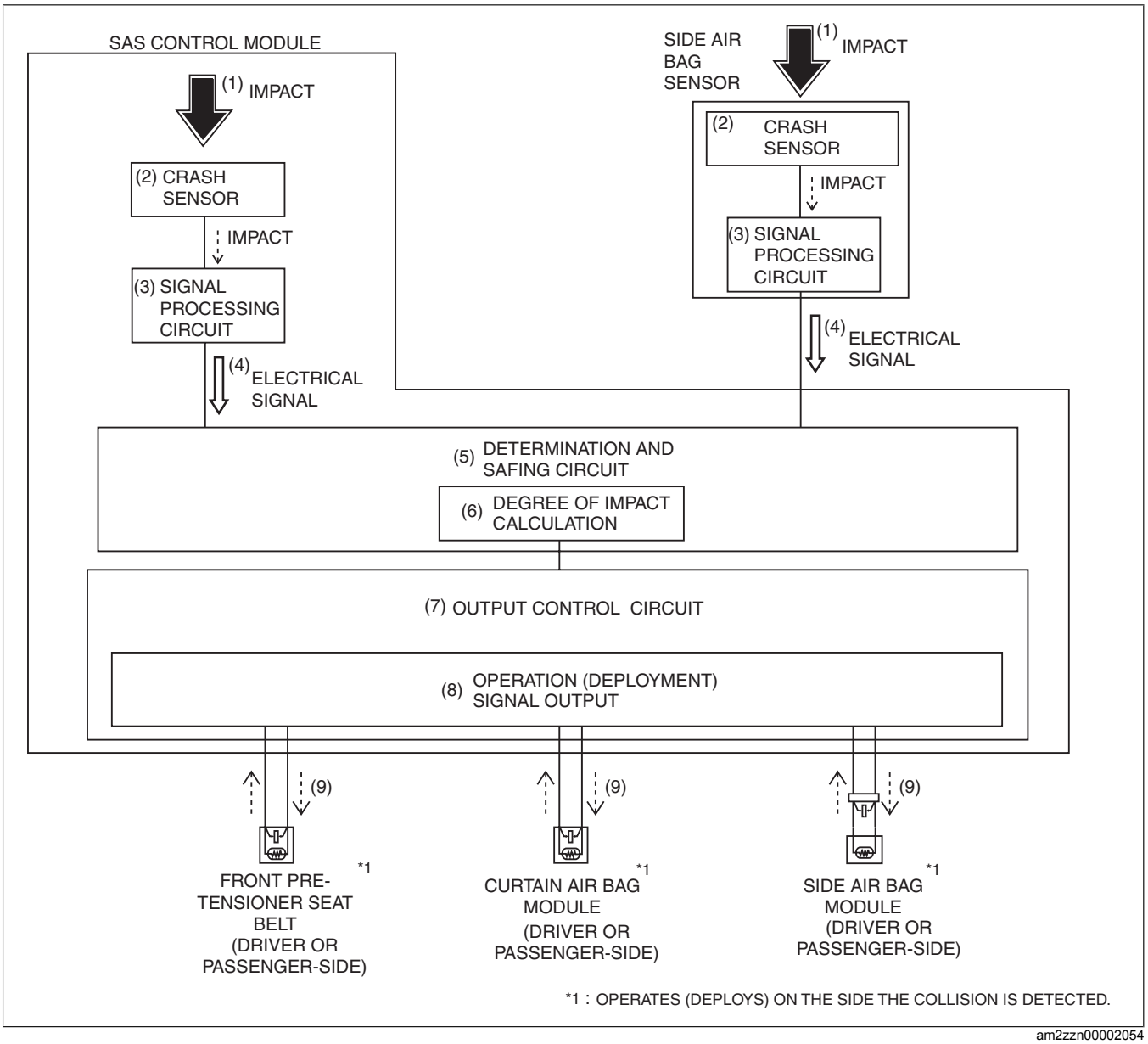
1. When an impact from a frontal or frontal offset collision (1) is received, the crash sensor built into the crash zone sensor and the crash sensor (2) built into the SAS control module detect the impact and convert the impact into an electrical signal at the signal processing circuit (3).
2. The SAS control module calculates (6) the degree of impact using the determination and safing circuit (5) based on the electric signal (4) from each crash sensor.
3. If the calculated degree of impact exceeds the previously set threshold, the SAS control module outputs (8) an operation (deployment) signal to each air bag module or pre-tensioner seat belt at the output control circuit (7).
4. Each air bag module or pre-tensioner seat belt operates (deploys) (9) based on the operation (deployment) signal from the SAS control module.



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Operation during a lateral collision (with side air bag module and curtain air bag module)

1. When an impact from a lateral collision (1) is received, the crash sensor built into the side air bag sensor and the crash sensor (2) built into the SAS control module detect the impact and convert the impact into an electrical signal at the signal processing circuit (3).
2. The SAS control module calculates (6) the degree of impact using the determination and safing circuit (5) based on the electric signal (4) from each crash sensor.
3. If the calculated degree of impact exceeds the previously set threshold, the SAS control module outputs (8) an operation (deployment) signal to each air bag module or pre-tensioner seat belt at the output control circuit (7).
4. Each air bag module or pre-tensioner seat belt operates (deploys) (9) based on the operation (deployment) signal from the SAS control module.



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